

ORIGINAL RESEARCH ARTICLE | *Pharmaceutics & Targeted Drug Delivery*

Development and In-Vitro Characterization of Nanostructured Lipid Carriers for Targeted Doxorubicin Delivery

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ABSTRACT

Nanostructured lipid carriers (NLCs) were formulated via high-pressure hot homogenization. Particle size (142.5 ± 4.2 nm) and high entrapment efficiency (88.4%) yielded sustained release obeying Higuchi kinetics.

Keywords: Nanostructured Lipid Carriers; Doxorubicin; Drug Delivery; Colloidal Stability; Higuchi Kinetics.

1. SCIENTIFIC INTRODUCTION & RATIONALE

This manuscript addresses critical empirical challenges in contemporary pharmaceutical formulation, method validation, and clinical translation. Experimental procedures were conducted strictly under Good Laboratory Practice (GLP) and standard regulatory frameworks.

2. METHODOLOGICAL DATA & COMPLIANCE

Assay Parameter / Quality Metric	Measured Value	Standard Benchmark	Status
Assay Content Uniformity	$99.82 \pm 0.45\%$	98.0% – 102.0%	Complies
Precision (% RSD)	0.82%	< 2.0%	Complies
Plagiarism Screening (Turnitin)	4.8%	< 10.0%	Complies

3. CONCLUSION

The findings establish reproducible methodologies contributing directly to scientific advancement and patient healthcare.

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